

Week 2

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1 Solving MDPs through Model-Free Methods (Reinforcement Learning)

When the environment is perfectly modeled, we possess exact knowledge of the Transition Probability $P(s'|s, a)$ and the Reward Function $R(s, a, s')$. In this theoretical setting, we can use **Dynamic Programming (DP)** to solve for the State-Value Function (V) or Action-Value Function (Q) functions. By iteratively applying the Bellman equations, we can compute the Policy (π) that yields the highest possible return without ever needing to interact with the environment.

But, In most real-world scenarios, the transition probabilities and reward functions are unknown. Here, we use **Reinforcement Learning (RL)** to estimate these values through trial and error.

1.1 temporal Differencing (TD)

Methods that learn by observing the differences between predicted rewards and actual rewards received after taking an action. There are primarily 2 temporal differencing methods used:

1.1.1 Q-Learning (Off-Policy)

The agent learns the value of the optimal policy regardless of the actions it is actually taking. It updates its $Q(s, a)$ estimates based on the maximum possible future reward.

1.1.2 SARSA (On-policy)

The agent learns the value of the policy it is currently following, including the exploratory moves it makes. It updates $Q(s, a)$ based on the action it actually takes next.

1.2 Monte Carlo Methods

These learn from complete episodes of experience by averaging the total returns received after visiting a state. Unlike TD learning, they only update after an episode ends.

1.3 Deep Actor-Critic Methods

While traditional TD methods (like Q-Learning) learn value functions to derive a policy, modern Deep RL often employs **Actor-Critic** architectures. These methods combine the benefits of policy-based learning (learning the strategy directly) and value-based learning (using a *critic* to evaluate actions via TD errors). Examples include:

- **PPO (Proximal Policy Optimization):** An on-policy method focused on stable, incremental policy updates.
- **DDPG (Deep Deterministic Policy Gradient):** An off-policy method designed for continuous action spaces.
- **SAC (Soft Actor-Critic):** An off-policy method that maximizes both reward and entropy for robust exploration.

Glossary

MDP Markov Decision Process. A mathematical framework used to model decision-making situations where outcomes are partly random and partly under the control of a decision-maker.

State Space (S) The set of all valid situations, configurations, or positions ($s \in S$) that the agent can occupy within the environment.

Action Space (A) The set of all possible moves, choices, or decisions ($a \in A$) available to the agent in its current situation.

Transition Probability (P) The probability $P(s' | s, a)$ that executing a specific action a in state s will cause the environment to shift to a specific next state s' . This accounts for natural randomness or noise in the environment.

Reward Function (R) The immediate numerical feedback $R(s, a, s')$ given to the agent by the environment right after moving from state s to state s' via action a .

Policy (π) The agent's strategic brain or rulebook. Formally written as $\pi(a | s)$, it maps a given state to the probability distribution of selecting each available action. The ultimate goal of the agent is to find the Optimal Policy (π^*).

State-Value Function (V) Denoted as $V(s)$, it measures how good it is to be in a specific state s by calculating the total expected future reward accumulated from that point onward, assuming the agent follows its current strategy until the end.

Action-Value Function (Q) The Quality function, denoted as $Q(s, a)$, which calculates the total expected long-term return if the agent commits to a specific initial action a from state s , and follows its policy thereafter.

Discount Factor A factor $\gamma \in [0, 1]$ that determines the present value of future rewards.

Transition Probability The probability $P(s' | s, a)$ of transitioning to state s' from state s after taking action a .

Reward Function The function $R(s, a, s')$ that gives the immediate reward for transitioning from state s to s' via action a .